Multimodal Deep Learning

Solutions to Exercise Sheet 1

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Exercise 1: Identifying Multi-Modal Pipelines

For each of the following applications, state which primary multi-modal pipeline type (Translation, Alignment, or Fusion) described in the lecture would be most appropriate. Provide a brief (1-2 sentence) justification for your choice based on the pipeline's core function.

(a) **Input:** A photograph of a menu written in Japanese.
Output: An itemized list of the menu items in English.

How would the pipeline choice change if the goal was:
Input: The same photograph of the Japanese menu and the text query "Does this menu contain shrimp?".
Output: A binary "Yes" or "No" answer.

(b) **Input:** The text description "a fluffy German shepherd dog catching a blue ball mid-air in a park" and a large database containing millions of diverse images.
Output: The index or identifier of a single image from the database that most closely matches the text description.

(c) **Input:** A 10-second video clip from a patient's echocardiogram and a table containing the patient's vital signs (age, heart rate, blood pressure).
Output: A binary classification indicating the presence of significant valve insufficiency.

(d) **Input:** A paragraph from a fantasy novel describing a dragon landing atop a castle tower during a storm.
Output: A generated image visualizing this specific scene.

Solution:

(a)
- **Pipeline:** Translation
- **Justification:** This task involves converting data directly from one modality (Image) into another modality (Text). The core function is transformation between modalities.

Follow-up Task (Menu Photo + Text Query -> Yes/No):

- **Pipeline:** Fusion
- **Justification:** This requires combining information from two distinct input modalities (Image and Text) to produce a single output prediction (binary classification). The core function is integration of multiple inputs.

(b)
- **Pipeline:** Alignment
- **Justification:** The goal is to find the best match between a text query and images in a database. This is typically achieved by projecting both text and images into a shared embedding space where distances correspond to semantic similarity, allowing retrieval. The core function is mapping different modalities to a common representation for comparison.

(c)
- **Pipeline:** Fusion
- **Justification:** Information from multiple input modalities (Video and Tabular data) needs to be integrated or combined to make a final prediction (binary classification). The core function involves merging multimodal inputs.

(d)
- **Pipeline:** Translation
- **Justification:** This task directly converts input data from one modality (Text) into a different output modality (Image). The core function is generation based on transformation between modalities.

Exercise 2: Intuition for Manifolds

Many real-world datasets, while represented in a high-dimensional space $\mathbb{R}^D$, often have inherent structure meaning the data points effectively lie on or near a lower-dimensional geometric shape embedded within that space.

For each case described below, reason about and identify:

i. the natural **data space** $\mathbb{R}^D$,

ii. the **data manifold** $M \subset \mathbb{R}^D$,

iii. its **intrinsic dimension** m, and

iv. a sensible **coordinate space** $U \subset \mathbb{R}^m$.

Consider the following cases:

1. **Robot arm.** A planar two-segment robot arm with a fixed base, allowing shoulder rotation and elbow bend. Analyze this system according to points (i)-(iv) above.

Solution:

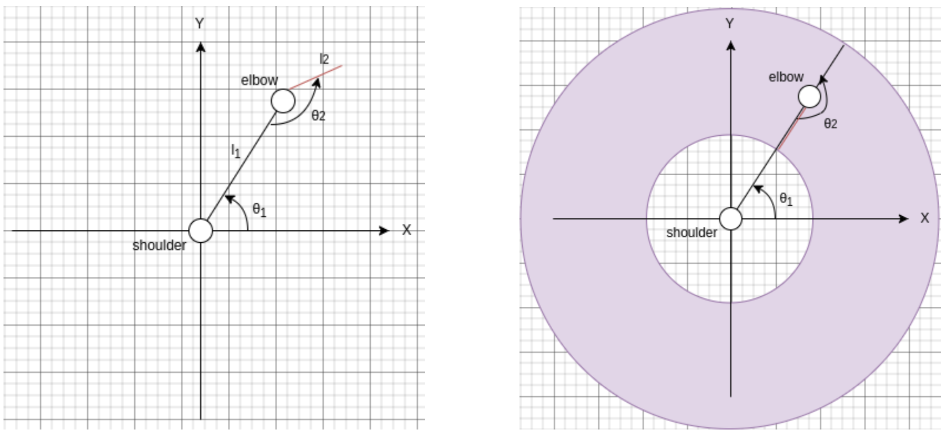

Figure 1: Sketch of the robotic arm (left) and its reachable workspace, an annulus (right).

$$x = \ell_1 \cos\theta_1 + \ell_2 \cos(\theta_1 + \theta_2)\,, \tag{1}$$

$$y = \ell_1 \sin\theta_1 + \ell_2 \sin(\theta_1 + \theta_2)\,. \tag{2}$$

The endpoint position (x, y) depends on the two angles θ_1 and θ_2.

i. **Data Space ($\mathbb{R}^D$):** The space of endpoint positions (x, y). Thus, $D = 2$, and the space is $\mathbb{R}^2$.

ii. **Data Manifold (M):** The set of all reachable (x, y) points. This forms a ring shape centered at the origin in $\mathbb{R}^2$. Specifically, points (x, y) such that $|L_1 - L_2| \leq \sqrt{x^2 + y^2} \leq L_1 + L_2$. M is this 2D region. Compare figure 1.

iii. **Intrinsic Dimension (m):** The configuration is fully determined by two independent parameters, θ_1 and θ_2. Therefore, $m = 2$.

iv. **Coordinate Space (U):** The space of the joint angles (θ_1, θ_2). For example $U = [0, \pi) \times [0, 2\pi) \subset \mathbb{R}^2$. Each point in U maps to a point in M.

Note: In this specific case, $m = 2$ equals the ambient dimension $D = 2$, but the manifold M is still a restricted subset of the ambient space $\mathbb{R}^2$.

2. **Hand-written digits (MNIST).** Each image is represented as a 784-dimensional vector (flattened 28x28 pixel image).

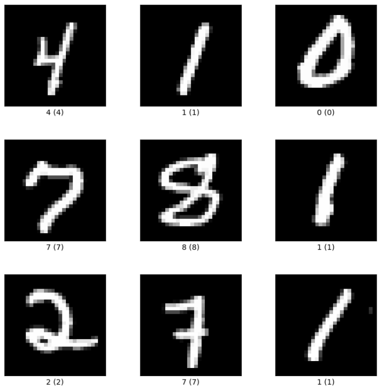

Figure 2: Examples of MNIST handwritten digits. [1]

First, analyze this data set according to points (i)-(iv) above.

Solution

i. **Data Space ($\mathbb{R}^D$):** Flattened images, $D = 784$. Space $\mathbb{R}^{784}$ (or subset $[0, 255]^{784}$).

ii. **Data Manifold (M):** Subset of $\mathbb{R}^{784}$ representing plausible handwritten digits.

iii. **Intrinsic Dimension (m):** Much smaller than 784 ($m \ll 784$), reflecting limited degrees of freedom in writing digits.

iv. **Coordinate Space (U):** Low-dimensional space $U \subset \mathbb{R}^m$, often a latent space learned by models (e.g., autoencoders) capturing variations like style

Then, address the following specific questions:

(a) Argue qualitatively why the intrinsic dimension m is expected to be much smaller than the ambient dimension $D = 784$ (i.e., $m \ll 784$).

(b) Suggest a type of learned coordinate space (e.g., from dimensionality reduction or generative models) that could potentially "unfold" this data manifold.

(c) Outline a method to approximate the geodesic distance (distance along the manifold) between two image points on the manifold M.

Solution

(a) Most vectors in $\mathbb{R}^{784}$ represent random noise, not valid digits. Valid digits exhibit strong structure: nearby pixels are correlated, strokes are continuous, there are constraints on shape/topology (e.g., loops in '8', endpoint counts in '3'), and variations in writing style (slant, thickness) are limited. These factors severely restrict the possible points in $\mathbb{R}^{784}$ that constitute the manifold M.

(b) Autoencoders learn an encoder $f_\theta : M \to U$ and a decoder $g_\phi : U \to M$ (or M'). The space U is the learned hidden feature space (or latent space). The goal is to find a space U where the data representation is disentangled, semantically meaningful, and potentially simpler. It represents an "unfolding" of the complex manifold M.

(c) Construct a k-Nearest Neighbor (k-NN) (small k keeps edges local to the manifold) graph where each MNIST image x_i is a node. Connect each node x_i to its k nearest neighbors in the ambient space $\mathbb{R}^{784}$ (using Euclidean distance $\|x_i - x_j\|_2$ as edge weights). Approximate the geodesic distance between x_1 and x_2 by finding the shortest path distance between their corresponding nodes in this graph (e.g., using Dijkstra's algorithm). The geodesic distance, approximated by the graph path, represents the shortest path along the manifold M (staying "close" to valid digits) and is generally greater than or equal to the Euclidean distance. It often better reflects perceptual or semantic similarity for small changes along the manifold. (strategy used by Isomap)

Exercise 3: Distance in Manifolds

The distance metric used can significantly impact interpretation, especially when dealing with manifolds. Consider the *Euclidean distance* (straight-line in the Data Space $\mathbb{R}^D$) and the *Geodesic distance* (shortest path along the Data Manifold M).

For each scenario, compare the Euclidean and geodesic distance between two points A and B:

(a) **Earth.** A = London and B = Sydney.

(b) **Knotted rope.** Two points on a tangled rope lying on a table.

You can use the concept of a nearest neighbor graph to illustrate why geodesic distance can differ significantly from Euclidean distance on a manifold.

Solution

(a) **Earth's Surface (London to Sydney):**

- **Comparison:** The geodesic distance (shortest path along the Earth's curved surface, the great-circle distance) is significantly *larger* than the Euclidean distance (the straight line tunnel through the Earth's interior).
- **Reason:** The Euclidean path takes a shortcut through the ambient space ($\mathbb{R}^3$) that is not available if constrained to the manifold (Earth's surface $M \approx S^2$).

(b) **Knotted rope on a table**

- **Comparison:** Geodesic distance d_G (along the rope) can be vastly *larger* than Euclidean distance d_E (straight across the table).

- **Reason:** The rope's embedding folds the 1D manifold M. Points close in the ambient $\mathbb{R}^2$ can be far apart along the rope itself. E.g., $d_E = 2$ cm vs $d_G = 150$ cm.

A k-NN graph connects points that are close in the ambient space. By finding the shortest path within this graph, we chain together short Euclidean segments that are likely to lie close to the manifold. This sequence of short hops approximates the curved path along the manifold (geodesic distance), preventing large shortcuts through the ambient space that violate the manifold constraint, especially if the manifold is curved or has complex topology.

Exercise 4: Topology and Classification

The lecture discussed how deep learning involves transformations that aim to unfold complex data manifolds into simpler representations, often making non-linearly separable data linearly separable in a hidden feature space. In this exercise, you will implement and visualize this process using a simple dataset and a Multi-Layer Perceptron (MLP).

Find instructions in the accompanying file: 'exercise1.ipynb'.

References

[1] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," in *Proceedings of the IEEE*, vol. 86, pp. 2278–2324, IEEE, 1998.